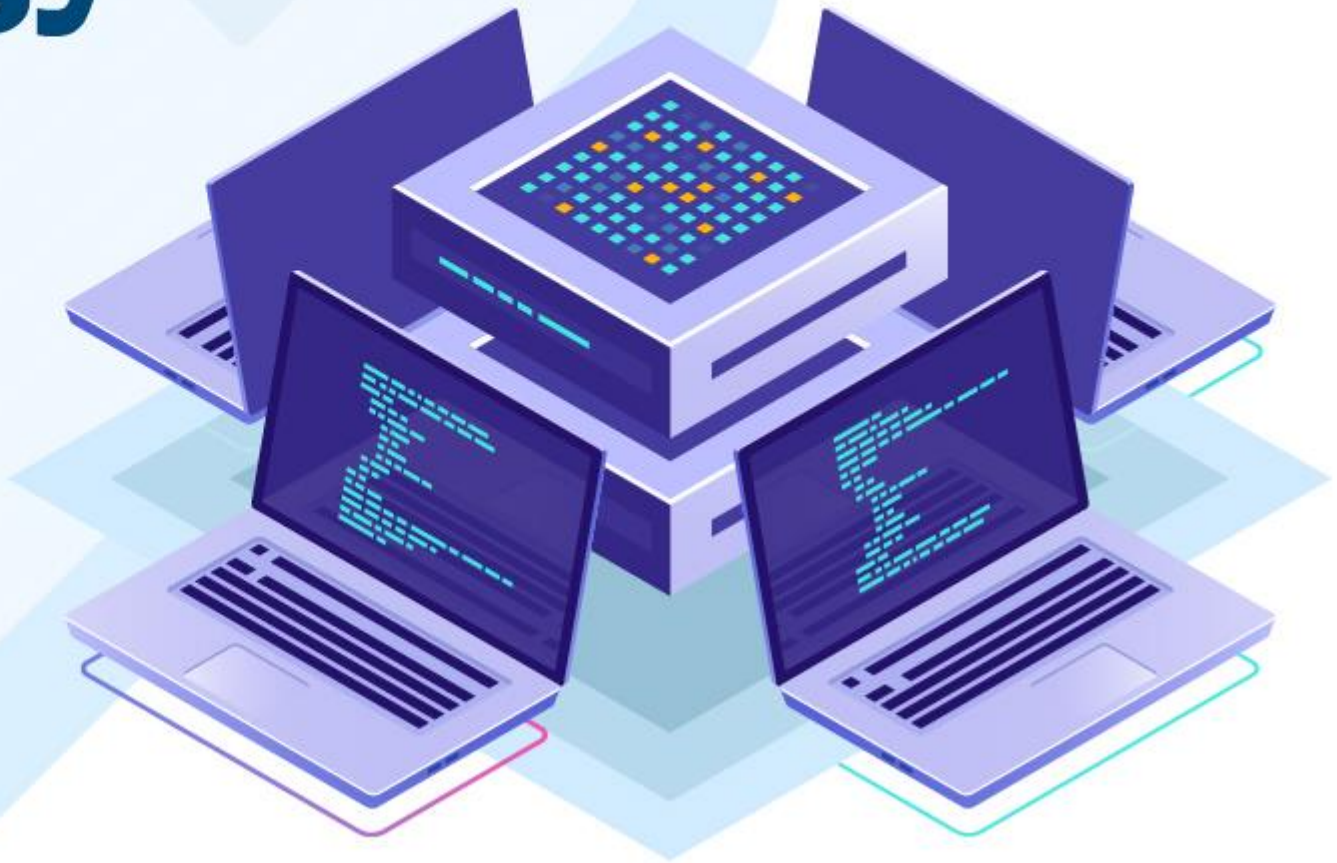
Information Technology

Fall Semester
2026 - 2027

- Embedded Systems
- **Dr. Yasmine Mahmoud**



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Student assessments + Assessment Grades

10 Project

10 Quizzes (2 quizzes each 5) – FTF/Lectures

5 Assignment + 5 Section Attendance

20 Midterm Exam

50 Final Exam

Embedded Systems

TEXT BOOKS:

1. Introduction to Embedded Systems - shibu k v, Mc Graw Hill Education.
2. Computers as Components –Wayne Wolf, Morgan Kaufmann (second edition).

REFERENCE BOOKS:

1. Embedded System Design -frank vahid, tony grivargis, john Wiley.
2. Embedded Systems- An integrated approach - Lyla b das, Pearson education 2012.
3. Embedded Systems – Raj Kamal, TMH

1. Introduction to Embedded Systems

What is Embedded System (ES)?

- An Electronic/Electro mechanical system which is designed to perform a specific function and is a combination of both hardware and firmware (Software).
- E.g. Electronic Toys, Mobile Handsets, Washing Machines, Air Conditioners, Oven, DVD Player etc…





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1. Introduction to Embedded Systems

What is Embedded System (ES)?

- An embedded system is a combination of 3 things:
  a) Hardware
  b) Software
  c) Mechanical Component

And it is supposed to do one specific task only.

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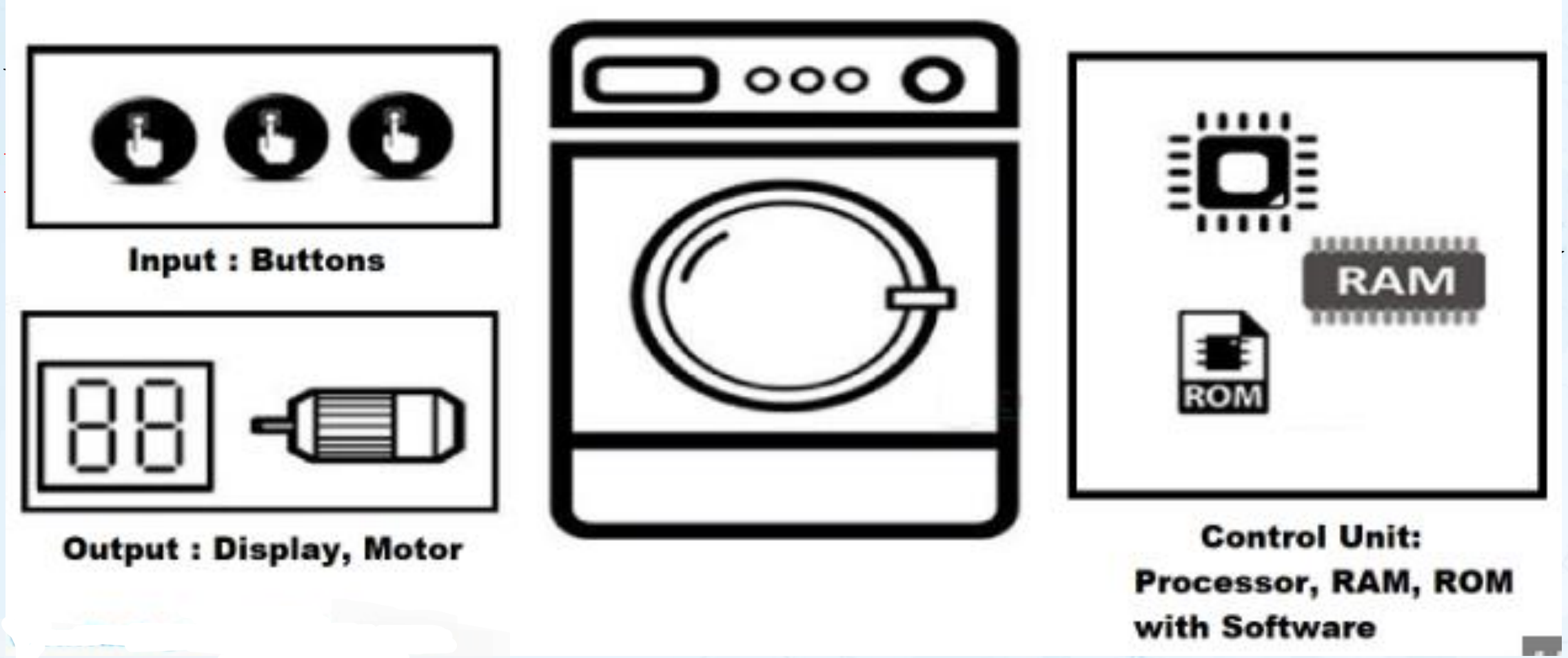
1. Introduction to Embedded Systems

What is Embedded System (ES)?

Example 1: Washing Machine

- A washing machine from an embedded systems point of view has:
  a) Hardware: Buttons, Display & buzzer, electronic circuitry.
  b) Software: It has a chip on the circuit that holds the software which drives controls & monitors the various operations possible.
  c) Mechanical Components: the internals of a washing machine which actually wash the clothes control the input and output of water, the chassis itself.

1. Introduction to Embedded Systems



1. Introduction to Embedded Systems

What is Embedded System (ES)?

Example 2: Air Conditioner

- An Air Conditioner from an embedded systems point of view has:
  a) Hardware: Remote, Display & buzzer, Infrared Sensors, electronic circuitry.
  b) Software: It has a chip on the circuit that holds the software which drives controls & monitors the various operations possible. The software monitors the external temperature through the sensors and then releases the coolant or suppresses it.
  c) Mechanical Components: the internals of an air conditioner the motor, the chassis, the outlet, etc…

1. Introduction to Embedded Systems

What is Embedded System (ES)?

- An embedded system is designed to do a specific job **only**.
  - Example: a washing machine can only wash clothes; an air conditioner can control the temperature in the room in which it is placed.
- The hardware & mechanical components will **consist** all the physically visible things that are used for input, output, etc.
- An embedded system will always have a chip (either microprocessor or microcontroller) that has the code or software which drives the system.

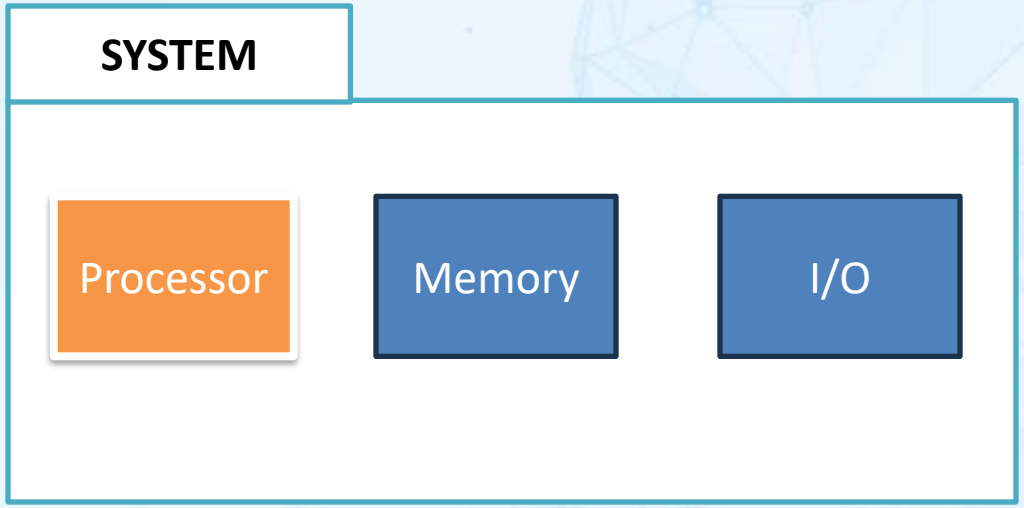
1. Introduction to Embedded Systems

What is Embedded System (ES)?

- Embedded Systems are:
  - ❑ Unique in character and behavior
  - ❑ With specialized hardware and software

Embedded Systems Concepts (ES Concepts)

- Any computing system must consist of **three** main things:
  - Processor
  - Memory
  - I/O



Embedded Systems Concepts (ES Concepts)

- Computing system is divided into **two** types:



Embedded Systems Concepts (ES Concepts)

- The Embedded System and the general-purpose computer are at two extremes.
- The embedded system is designed to perform a specific task whereas as per definition the general-purpose computer is meant for general use. It can be used for playing games, watching movies, creating software, work on documents or spreadsheets, etc…

Embedded Systems Concepts (ES Concepts)

- Computing system is divided into **two** types:
  1. **General purpose:** like our laptops, we can install many application, each application can make function different than others. But all applications are installed in the same resources (Same CPU, RAM, Same Hardware).

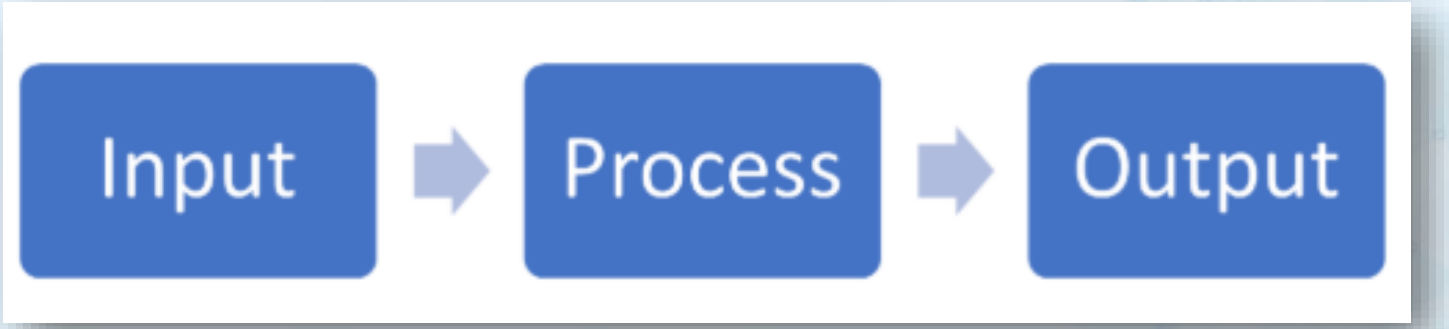


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Embedded Systems Concepts (ES Concepts)

- Computing system is divided into two types:

  2. **Specific purpose:** system is used for single purpose. Like fridge, microwave, etc.







Embedded Systems Vs General Computing Systems:

Criteria	General Purpose Computer	Embedded system
Contents	It is combination of generic hardware and a general purpose OS for executing a variety of applications.	It is combination of special purpose hardware and embedded OS for executing specific set of applications
Operating System	It contains general purpose operating system	It may or may not contain operating system.
Alterations	Applications are alterable by the user.	Applications are non-alterable by the user.
Key factor	Performance" is key factor.	Application specific requirements are key factors.
Power Consumption	More	Less
Response Time	Not Critical	Critical for some applications

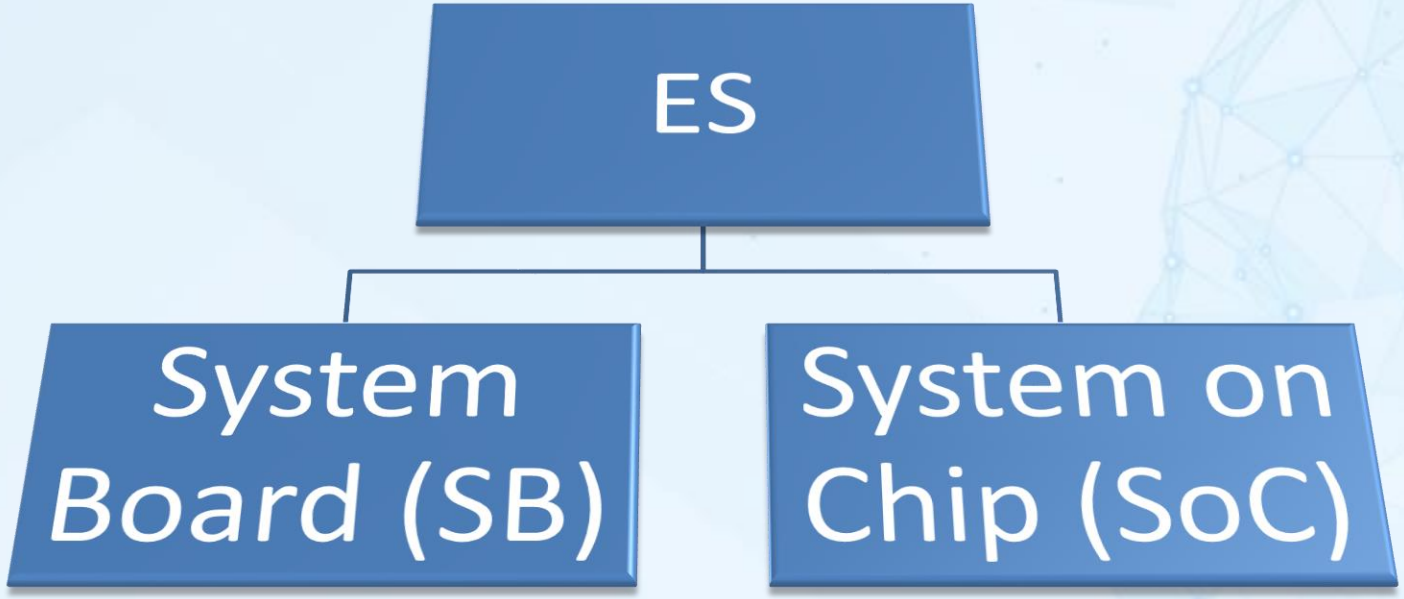
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Embedded Systems Concepts (ES Concepts)

- So embedded systems (Specific purpose system), they must have constrains.
- Where?
  - In power.
  - In cost.
  - In speed / time.
  - In size.
- So, while making embedded system, (power, cost, speed, and size) are **embedded system challenges**.

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How to make ES?

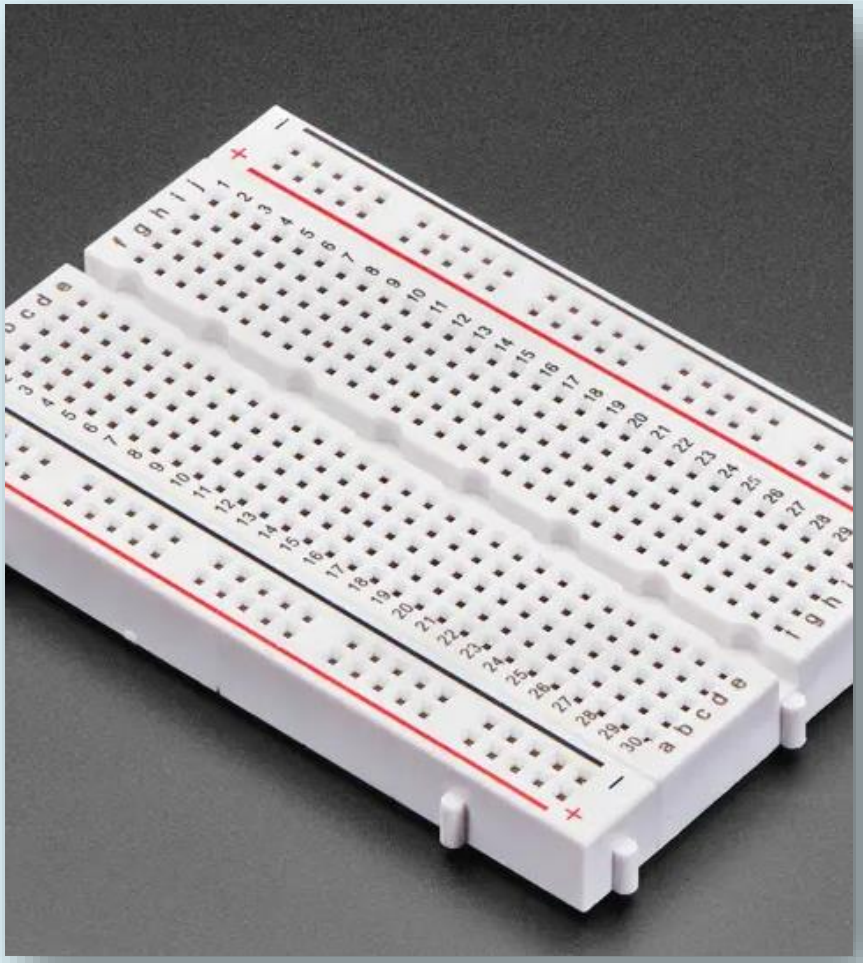


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How to make ES?

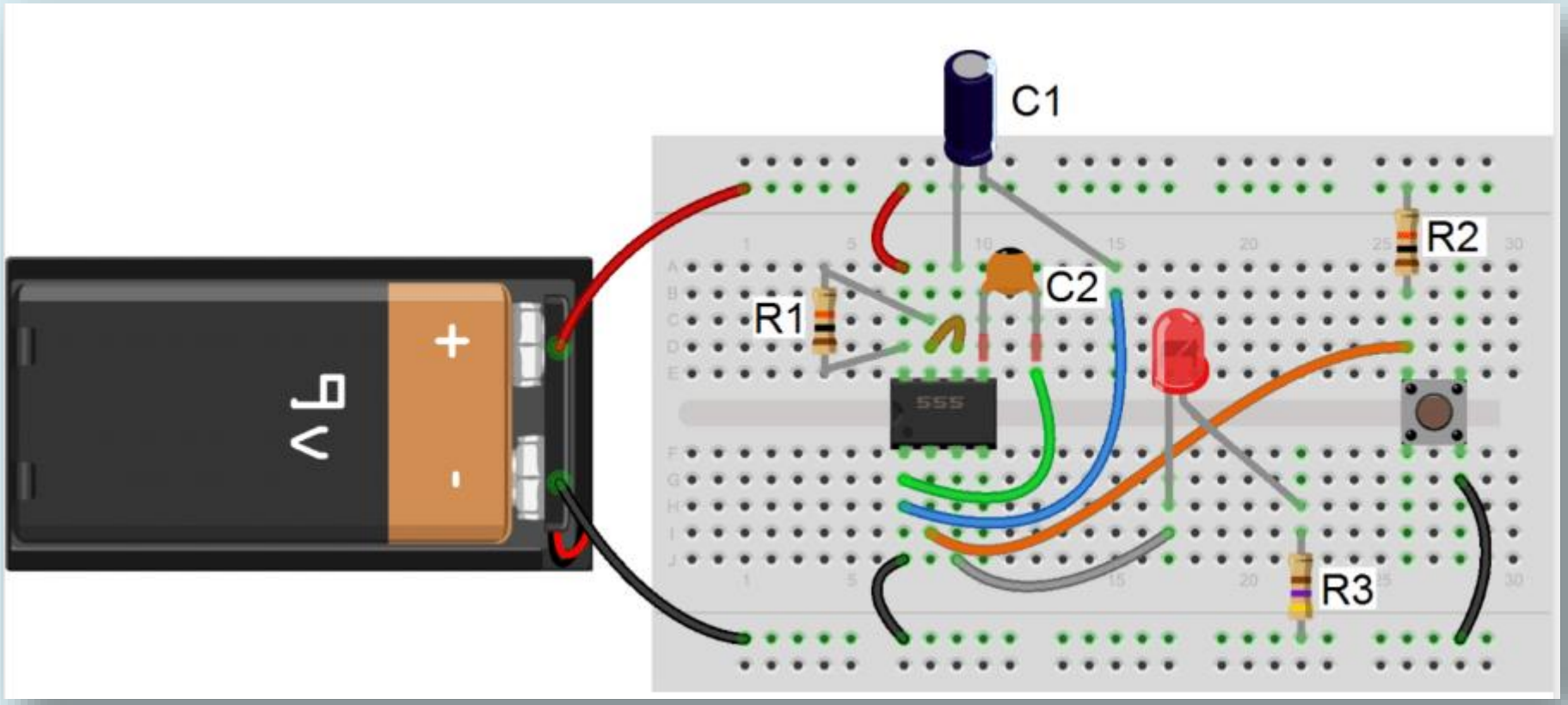
1. System board (SB): we can collect system on our breadboard.

- What is breadboard?
  - It is a board used to connect electronic components, such as wires, and processor, memory, and I/O preferences, to conduct various projects.



How to make ES?

1. System board (SB)



How to make ES?

2. System on chip (SoC):

- The chip is IC.
- Integrated circuit (IC) combines many or all high-level function elements of an electronic device onto a single chip instead of using separate components mounted to a motherboard, as is done in traditional electronics design.



Comparison Based on ES Challenges

points	SB	SoC
Size		
Cost		
Power		
Performance / Execution time		

Comparison Based on ES Challenges

points	SB	SoC
Size	Larger	Smaller
Cost	Expensive	Less Expensive
Power	Connect with wires that Consume power (High)	Less
Performance / Execution time	Same Performance	Same Performance

Comparison Based on ES Challenges

- Based on challenges, if we add configurability, so which one can be editable (applicable)?

points	SB	SoC
Configurability		

Comparison Based on ES Challenges

- Based on challenges, if we add configurability, so SB can be editable (applicable).

points	SB	SoC
Configurability	Yes	No

- So, in system's design phase, SB can be editable, if we want to make changes while designing the system.

Embedded Systems Concepts (ES Concepts)

IC (Integrated Circuit): It is a single chip.

Ex.

– Timer 555

- The 555 timer IC is an integrated circuit used in a many apps. of timer, delay, and pulse generation applications.
- It is one of the most popular timing ICs due to its flexibility and price.

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VLSI

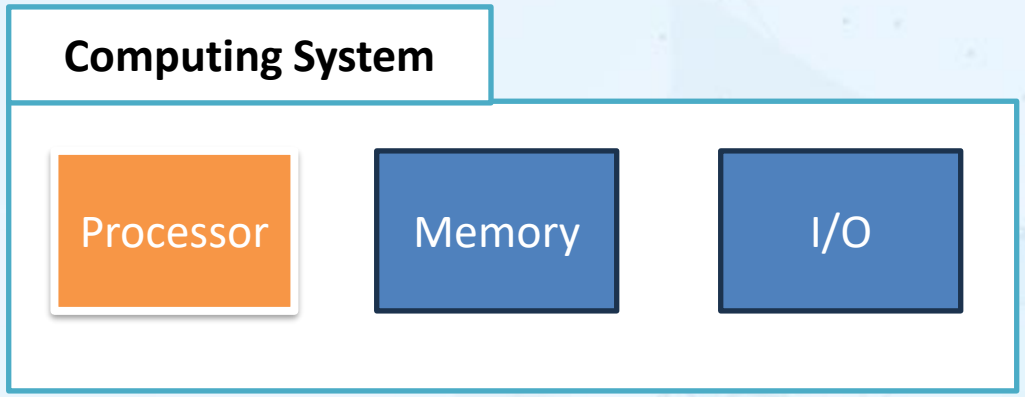
- Very Large-Scale Integrated Circuit (VLSI).
- VLSI is IC but with large numbers of gates; millions of transistors, millions of NAND gates. All inside the same size of IC.
- VLSI Benefits:
  - Small size.
  - Wide range functionality based on the wide range of NAND gates and transistors.

IC

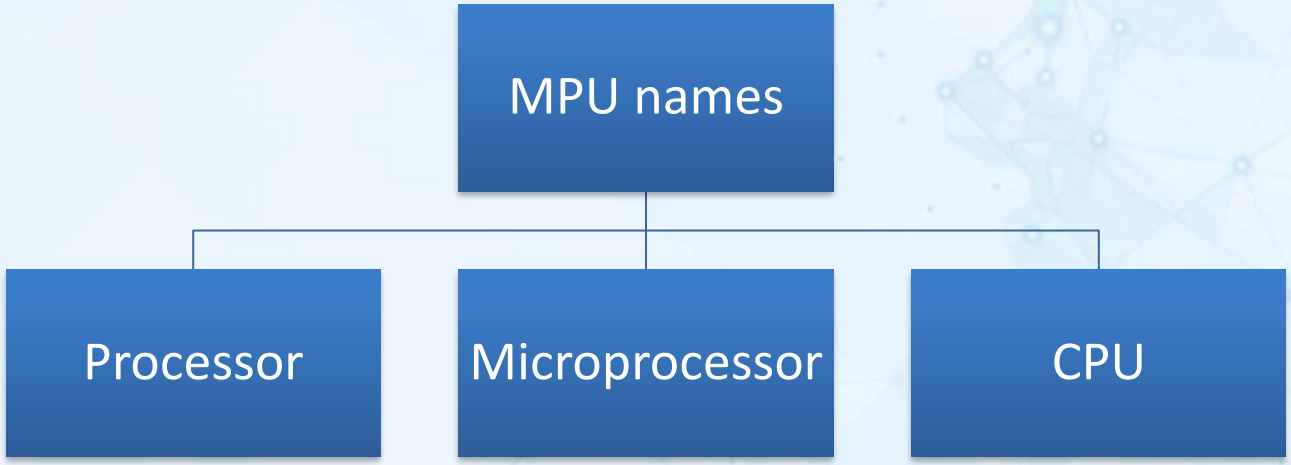
- So, Microprocessors, Microcontrollers, SoC, RAMs, Rom are **ICs**.

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MPU



- Microprocessor Unit (MPU): is a processor in the computing system.



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MPU

Processor
(old processors)



Each one is to represent one bit (0 or 1)
1 if tube is charged
0 otherwise
So, to make computer system we need a room

MPU



Microprocessor
(New Processors)

After manufacturing evolution, tubes are not used, but transistors.
Transistor is smaller than vacuum tubes.

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MPU



Microprocessor
(New Processors)

Also, we can make micro-things.
So, they called it microprocessor.

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MPU



CPU

(Primary / Master processor)

Central Processing Unit (CPU)

It is the master processor, and other microprocessors are slaves (GPU, DSP).

- CPU: is the primary processor
- GPU & DSP: are the secondary processors

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MPU



CPU

(Primary / Master processor)

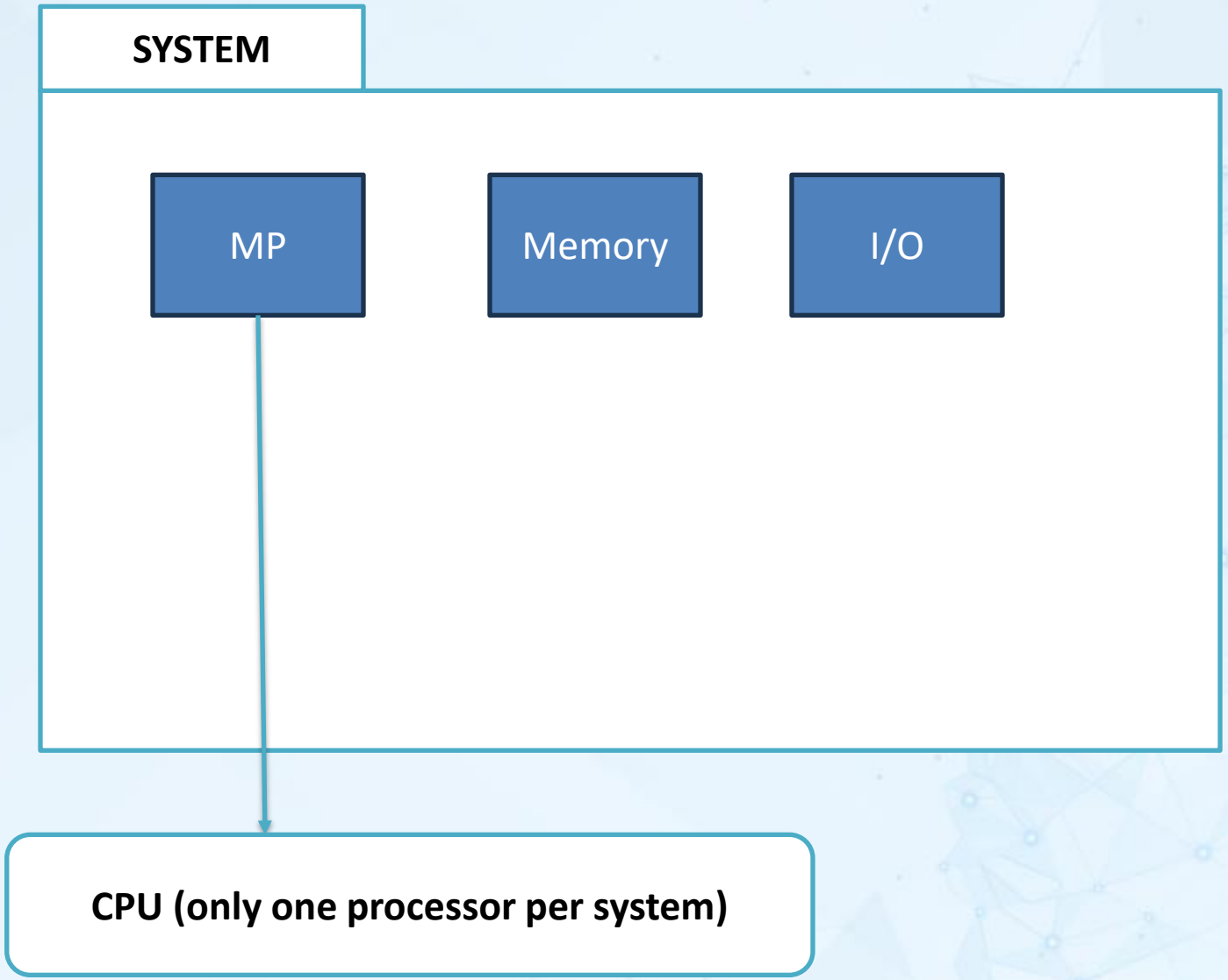
The system can have one CPU.
Otherwise, if the system have many microprocessors, so the primary one is the CPU.

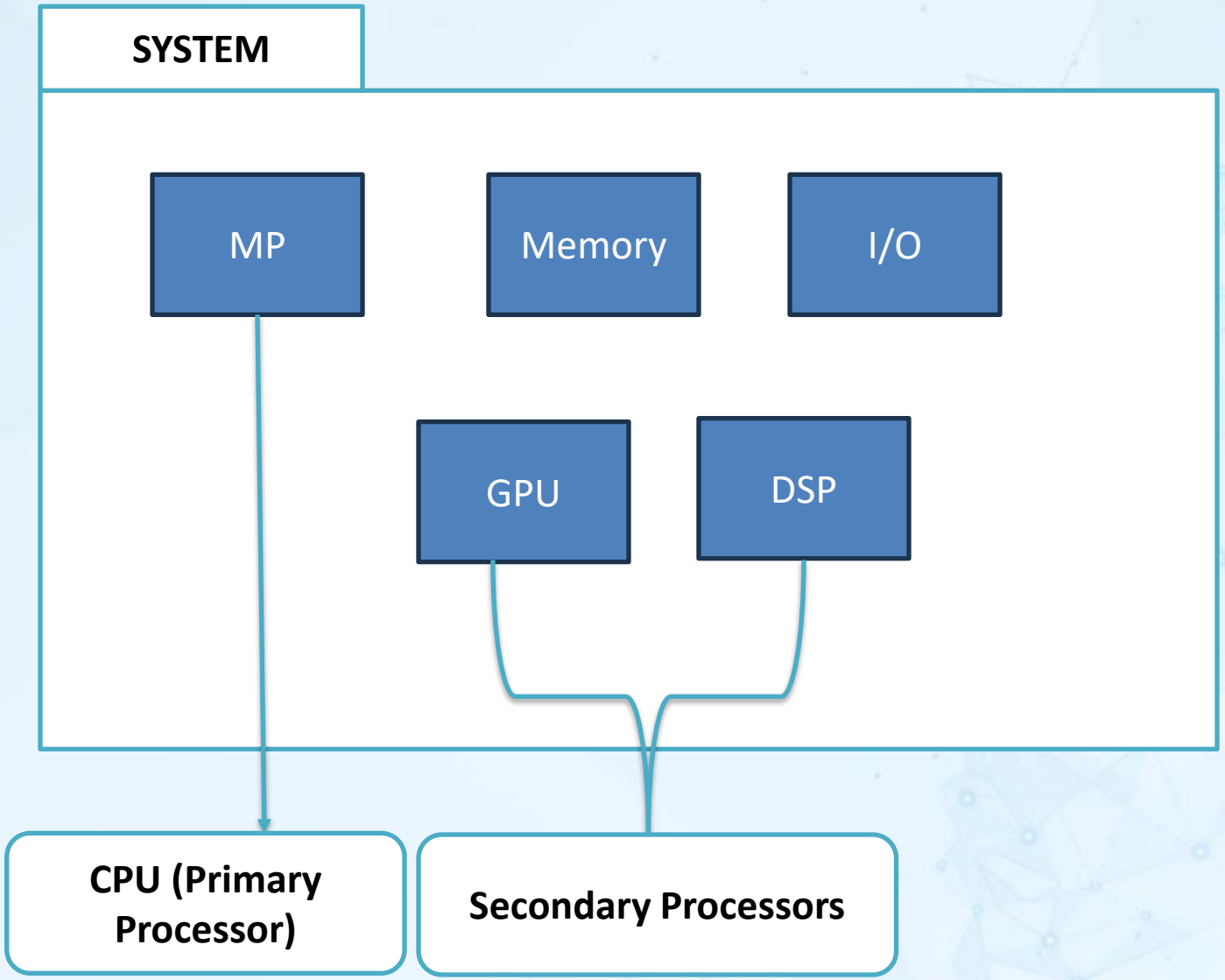
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MPU Vs. MCU

- MPU: Microprocessor Unit
- MCU: Microcontroller Unit
- MCU is a small computer on the same IC (single IC / Single chip)
- MCU is a computing system, which consists of three things:
  - Processor (MPU)
  - Memory
  - I/O
- So MPU **is a part of** MCU.

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SYSTEM
MP
Memory
I/O
CPU (only one processor per system)


SYSTEM
MP
Memory
I/O
GPU
DSP
CPU (Primary Processor)
Secondary Processors

EELU

GPU & DSP

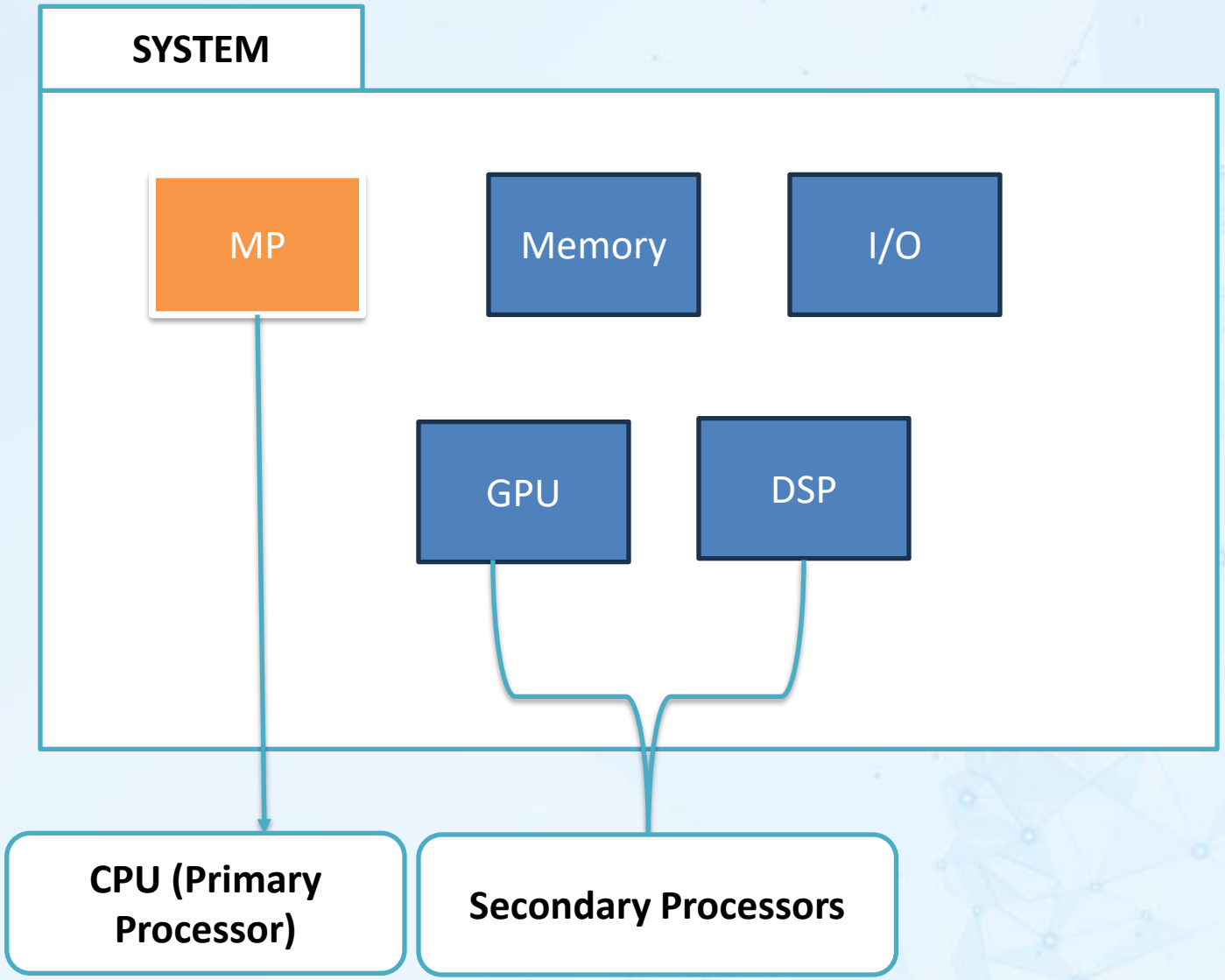
- Graphics Processing Unit (GPU) is a microprocessor.
- GPU understand the instructions that related to graphics.
- So, when MP (primary) want to do any operation related to graphics, will pass it to GPU (secondary processor).
- GPU assist MP in graphics operations. (apps. Like deep learning, graphics apps.)
- GPU also occasionally called Visual Processing Unit or VPU.

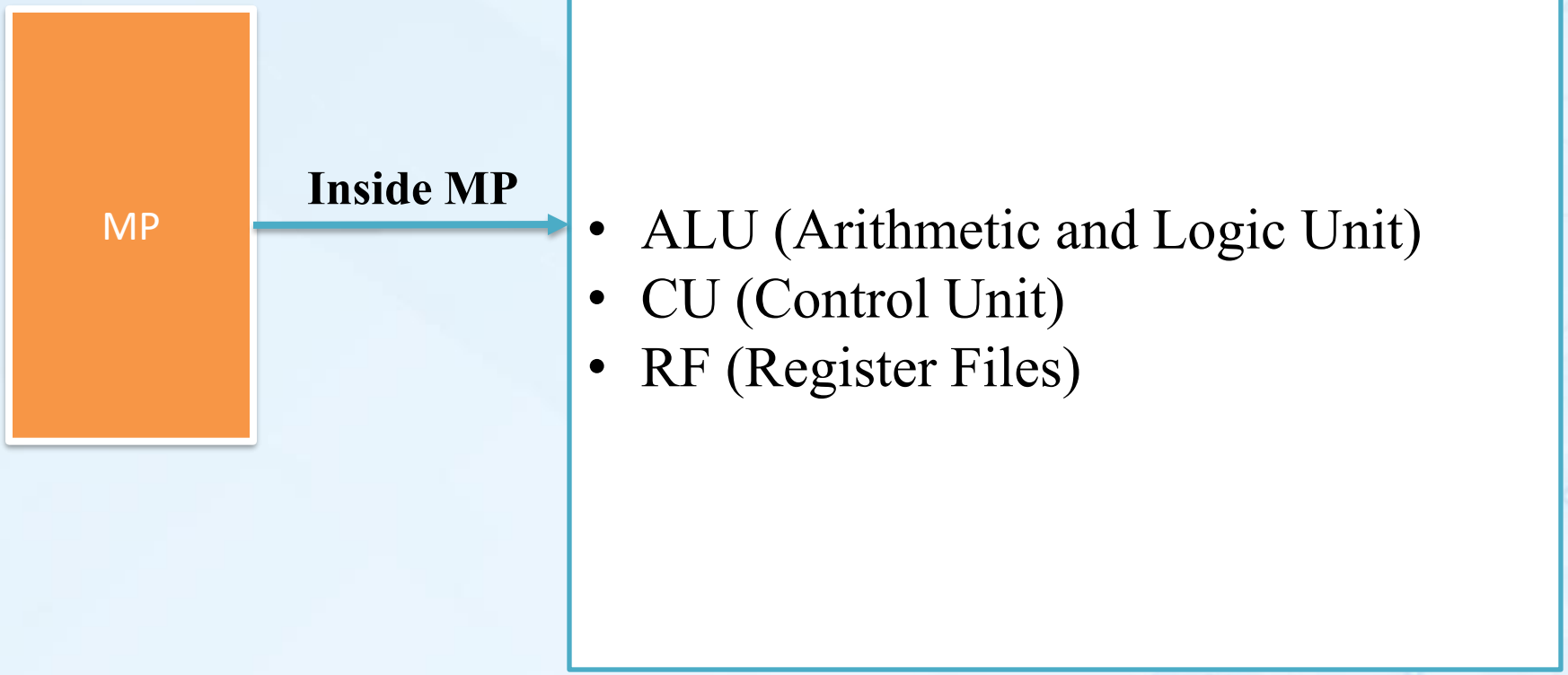
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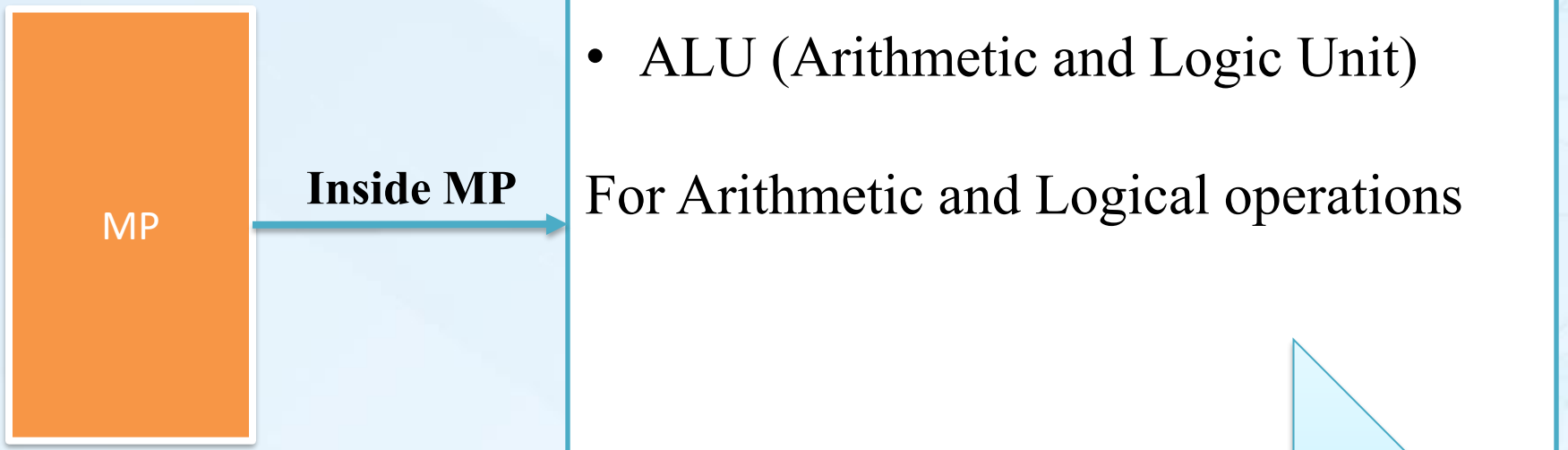
GPU & DSP

- Digital Signal Processor (DSP) is a microprocessor.
- DSP deals with complex computations.

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SYSTEM
MP
Memory
I/O
GPU
DSP
CPU (Primary Processor)
Secondary Processors


MP
Inside MP
• ALU (Arithmetic and Logic Unit)
• CU (Control Unit)
• RF (Register Files)



- But, for complex computations like (Fourier transform), we make secondary microprocessor (DSP) to deal with the computations.
- It decrease load on the primary processor.

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THANK YOU FOR WATCHING

QUESTIONS?

